Q3.

a) The investor solves $\max_{\omega_t} E_t[U(W_{t+1})]$ s.t. $W_{t+1} = W_t(R_{f,t+1} + \omega_t' x R_{t+1})$

Assume that $W_t > 0$, $W_{t+1} > 0$ almost surely, and that the optimal portfolio choice is interior. Under the usual regularity conditions, we can derive the first-order condition by differentiating expected utility w.r.t. ω_t

This problem can be written as $\max_{\omega_t} E_t[U(W_t(R_{f,t+1} + \omega_t' x R_{t+1}))]$

$$\frac{\partial E_t[U(W_t(R_{f,t+1} + \omega_t' x R_{t+1}))]}{\partial \omega_t} = U'(W_{t+1}) \cdot W_t \cdot x R_{t+1}$$

$$U(W) = \frac{W^{1-\gamma}}{1-\gamma}$$

$$U'(W) = W^{-\gamma}$$

Thus, $U'(W_{t+1}) = W_{t+1}^{-\gamma}$

$$\frac{\partial E_t[U(W_t(R_{f,t+1} + \omega_t' x R_{t+1}))]}{\partial \omega_t} = E_t[U'(W_{t+1}) \cdot W_t] = E_t[W_{t+1}^{-\gamma} \cdot W_t \cdot x R_{t+1}] = 0$$

$$E_t\left[\frac{W_t}{W_{t+1}^{\gamma}} x R_{t+1}\right] = W_t E_t\left[\frac{1}{W_{t+1}^{\gamma}} x R_{t+1}\right] = 0$$

Since $W_{t+1}^{\gamma} = (W_t \cdot R_{W,t+1})^{\gamma}$

$$E_t\left[\frac{1}{(W_t \cdot R_{W,t+1})^{\gamma}} x R_{t+1}\right] = W_t^{-\gamma} E_t[R_{W,t+1}^{-\gamma} x R_{t+1}] = 0$$

Since $W_t > 0$, we have

$$E_t[R_{W,t+1}^{-\gamma} x R_{t+1}] = 0$$

b)